Additive Manufacturing Letters 19 (2026) 100409

Contents lists available at ScienceDirect

Additive Manufacturing Letters

journal homepage: www.elsevier.com/locate/addlet

Short Communication

Influence of spatter formation on interfacial material mixing in multi-material laser powder bed fusion via *in situ* synchrotron x-ray radiation

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ARTICLE INFO

Keywords:
Additive Manufacturing
Multi-Material Laser Powder Bed fusion
Melt pool dynamics
Spatter formation

ABSTRACT

The goal of this work is to understand melt pool dynamics and material mixing in Multi-Material Laser Powder Bed Fusion (MM-LPBF), with particular emphasis on the influence of spatter formation at the distinct-material interface. The ability to selectively process steel-copper materials with finer resolution and greater design complexity in LPBF is of significant interest to the aerospace, nuclear, and structural industries because it enables the fabrication of components with both high strength and thermal performance. However, the understanding of melt-pool dynamics near interfacial regions in MM-LPBF remains limited, affecting the mixing zone and ultimately the interface properties. In this work, an *in-situ* X-ray synchrotron operando setup is used to investigate the melt pool evolution during MM-LPBF of bronze (Cu10Sn) and stainless steel 316 L. Computational fluid dynamics simulations complement the X-ray observations by analyzing thermal evolution, laser-metal interactions, and material mixing. The results show that spatter ejection disrupts melt pool stability, producing heterogeneous mixing characterized by elongated, abrupt islands in the interfacial region. Unlike single-material LPBF, MM-LPBF with dissimilar alloys that have markedly different thermal properties and laser energy absorptivity results in non-steady melt-pool dynamics that promote process instabilities and spatter formation. This work is the first to investigate the spatter formation mechanism in MM-LPBF using *in situ* X-ray analysis and provides new insights into process limitations by demonstrating that the standard laser beam profile is insufficient to effectively control material mixing at the interface.

1. Introduction

Laser Powder Bed Fusion (LPBF) is one of the most adapted industrial metal additive manufacturing processes for its ability to produce high-resolution part geometries, leverage part consolidation, and incorporate topology optimization for high-performance manufacturing. For instance, several reports on LPBF have noted 45% weight reduction in structural parts [1] and 50% improvement in heat sink performance [2], among other benefits. However, the current application of LPBF is limited to a single material component, in contrast to the functionally graded material composition achieved through directed energy deposition [3] and recent developments in MM-LPBF [4]. While the motivation for MM-LPBF [5] can be attributed to the potential advantages of selective deposition of different material systems in powder bed fusion [6], recent studies have highlighted the challenges of achieving defect-free interfaces between multi- materials within a single component [7–9].

One of the most studied materials systems in MM-LPBF is steel and copper alloys [4,7,10–25]. Copper alloys provide high thermal and electrical conductivity, and high-strength, corrosion-resistant steels compensate for their lower mechanical strength, which is desired for

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https://doi.org/10.1016/j.addlet.2026.100409
Received 15 May 2026; Received in revised form 5 July 2026; Accepted 18 July 2026
Available online 28 July 2026
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applications in extreme environments such as hypersonic and marine environments. However, there is a knowledge gap in understanding laser-material interactions within the melt pool of such immiscible materials, which could affect interface properties. Commonly observed defects at MM-LPBF material interfaces include porosity and solidification cracking, which can be attributed to the contrast in thermal properties and to material mixing driven by Marangoni convection [4,10]. The interface in MM-LPBF has been reported to range from 30 – 300 microns depending on build orientation [4,12]. For example, copper-steel vertical interfaces (i.e., intra-layer) have shallow mixing regions. On the other hand, the laser melting sequence influences the interface width in the horizontal orientation (i.e., inter-layer), with wider mixing when copper is built on steel than when steel is built on copper, due to differences in the thermal and laser absorption properties of the material systems. It is important to note that across both interface orientations, isolated alloy islands have been reported within the mixing zone, and non-homogeneous mixing could have induced solidification cracking and keyhole defects [26–28]. Studies have reported that melt pool dynamics is primarily responsible for material mixing and defect formation [29,30]. While the structure-property relationship of defect formation in MM-LPBF has been reported, the process-structure relationship, i.e., melt pool dynamics in MM-LPBF, has not been studied, which forms the aim of this work.

Understanding laser-material interactions in LPBF has been a critical focus for accelerating its adoption in applications. Thermo-fluid simulations incorporating multi-phase flow, free-surface motion of fluid, powder-particle interactions, angle-dependent absorptivity, and recoil pressure are used to study melt pool evolution [26,31–36]. It has been effective in understanding melt pool morphology, keyhole formation and evolution, and critical defects, such as lack of fusion, porosity, and residual stress [37,8]. On the other hand, *in-situ* high-energy X-ray radiation provides experimental insights and validation of simulation of processing physics that are otherwise impossible to observe in real time, i.e., sub-surface-to-scan-layer [38–41]. Several studies detail the mechanisms of keyhole phenomena [35,38], Marangoni convection effects due to thermal gradients [42,43], and spatter formation [36,44]. *In-situ* X-ray radiation data is also shown to be a viable option to inform 'ground truth' data for implementing *in-situ* process control for defect detection using a physics-based machine learning model [45,46].

Spatter generation is a critical phenomenon in AM processes, as it significantly influences melt pool dynamics. Two main types of spatter are reported in single-material LPBF processes: droplet spatter and powder spatter [44]. A small-spot-size laser produces intense irradiation, which rapidly raises the metal temperature at the focal point, leading to metal vaporization. Temperature-gradient-driven Marangoni flow, combined with metal vapor pressure, can destabilize molten metal and expel it as spatter particles, also called droplet spatter. The ejected molten jets subsequently undergo balling due to the Plateau-Rayleigh instability, minimizing surface energy by forming near-spherical liquid spatter particles [32]. The interaction with the vapor jet and other gas flows causes the jet to break up into smaller droplets [44]. The metallic vapor can also eject unmelted powder from the front and sides of the melt track, which is referred to as powder spatter. Spatter ejections further destabilize the melt pool by reducing the interacting metal volume, increasing the likelihood of keyhole collapse [47]. Spatter particles can further shadow the laser beam, reducing the effective input laser energy to the melt pool [48], and could result in unmelted powder and lack of fusion defects [34]. Other studies have shown that spatter formation is highly dependent on the AM machine architecture that controls the gas flows in the build chamber [29]. While the detrimental effects of melt pool instability and spattering have been investigated for single-material LPBF processes, this effect has not been investigated in MM-LPBF.

The objective of this work is to understand the melt-pool dynamics and effective material mixing in MM-LPBF, with a particular emphasis on the influence of the spatter-formation mechanism at the distinct-material interface (i.e., bronze Cu10Sn and stainless steel 316 L in this study). A distinct material interface is produced by scanning a single-layer laser track of the bronze (Cu10Sn) powder spread onto a steel plate (stainless steel 316 L). The experiment was performed within an *in-situ* X-ray synchrotron operando setup to record the melt pool evolution. A CFD simulation was used to investigate the process dynamics in detail, with a focus on material mixing. Dynamic melt pool behavior can be observed *in situ* during a spatter event, resulting in melt pool instability and backfilling. The simulation results match the *in-situ* X-ray observations and explain the nonhomogeneous mixing of the two materials observed in the sample. The observations and results are discussed to gain a fundamental understanding of process physics and better control the MM-LPBF process (Fig. 1).

2. Methodology

2.1. MM-LPBF experimental setup within an in-situ X-ray synchrotron operando setup

MM-LPBF experiments were conducted at the European Synchrotron Radiation Facility in a miniSLM setup developed by the Paul Scherrer Institute (PSI), Switzerland, which consists of an enclosed printing chamber, an adjustable powder-bed platform, a motorized, laser-equipped machine bed, and a powder delivery system [49]. An electrostatic powder spreader was used to deposit a uniform layer of metal powder [50]. As shown in Fig. 2a, the experimental setup included a stainless steel 316 L substrate (10 × 5 × 0.5 mm), gas-atomized Cu10Sn powder (D10 = 38 μm, D50 = 49 μm, and D90 = 61 μm) with a layer thickness of 100 μm, a laser scan track of 2 mm with a laser power of 300 W, and a scan speed of 400 mm/s. The enclosed chamber was purged with Argon until the oxygen level was reduced to 0.2%. Post-AM samples were cut and mounted longitudinally for material mixing characterization via scanning electron microscopy (SEM) after sample preparation, which involved sequential grinding with 320- and 600-grit papers, followed by polishing with 9 μm, 3 μm, and 1 μm diamond suspensions.

During laser scanning, melt pool dynamics were captured using a high-energy X-ray beam at 120 keV, in conjunction with a indirect detector based on lens-coupling a high-speed camera (Photron FASTCAM SA-Z) at 20,000 images/s. The acquired radiographic data were initially recorded at a frame size of 900 × 300 pixels and subsequently cropped to a region of interest of 566 × 150 pixels to isolate melt pool events. The corresponding pixel resolution was 12.4×10^{-3} mm/pixel. Considering the spatial and temporal resolution of data acquisition, all feature size measurements are prone to $\pm 6.2 \times 10^{-3} mm$ uncertainty, velocity measurements are prone to $\pm 0.175 \frac{mm}{s}$ *uncertaintyuncertainty*. Each frame was preprocessed using Contrast Limited Adaptive Histogram Equalization (CLAHE), which enhances local image contrast more effectively than conventional global contrast enhancement techniques by independently processing smaller user-defined pixel regions and subsequently blending the enhanced regions to minimize tiling artifacts. Critical features, including the keyhole, melt surface, and spatter particles, were then manually extracted by imposing an edge boundary based on intensity variation and labeling them for detailed analysis of melt pool dynamics. Fig. 2B summarizes the overall data-processing workflow and representative processed results.

2.2. Numerical simulation setup

To complement the experimental study of the MM-LPBF process, computational fluid dynamics (CFD) simulations were performed using the commercial software Flow3D-AM (version 2025A). The built-in multi-material modeling framework was employed to simulate two distinct materials within a single computational domain, with bronze and stainless steel defined separately (material properties are provided

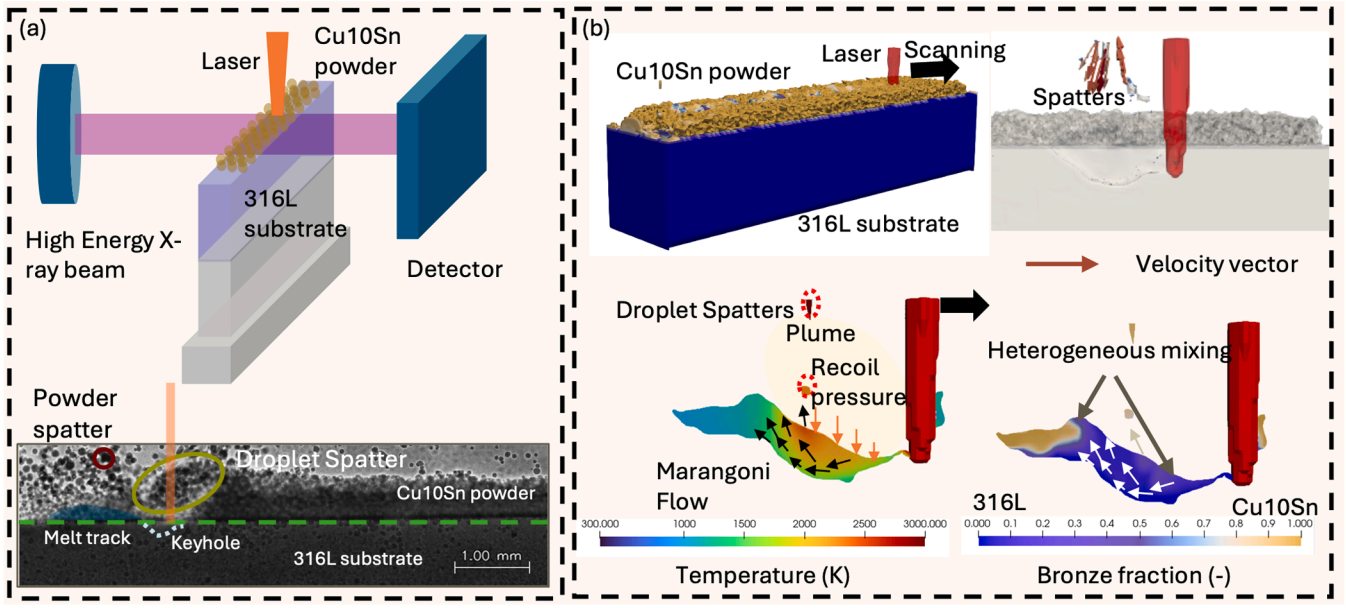


Fig. 1. (a) Schematic of *in-situ* X-ray synchrotron radiation set up to capture instability of MM-LPBF process with spattering event and (b) correlation to computation simulation and analysis of melt pool dynamics and melt flow induced material mixing in MM-LPBF.

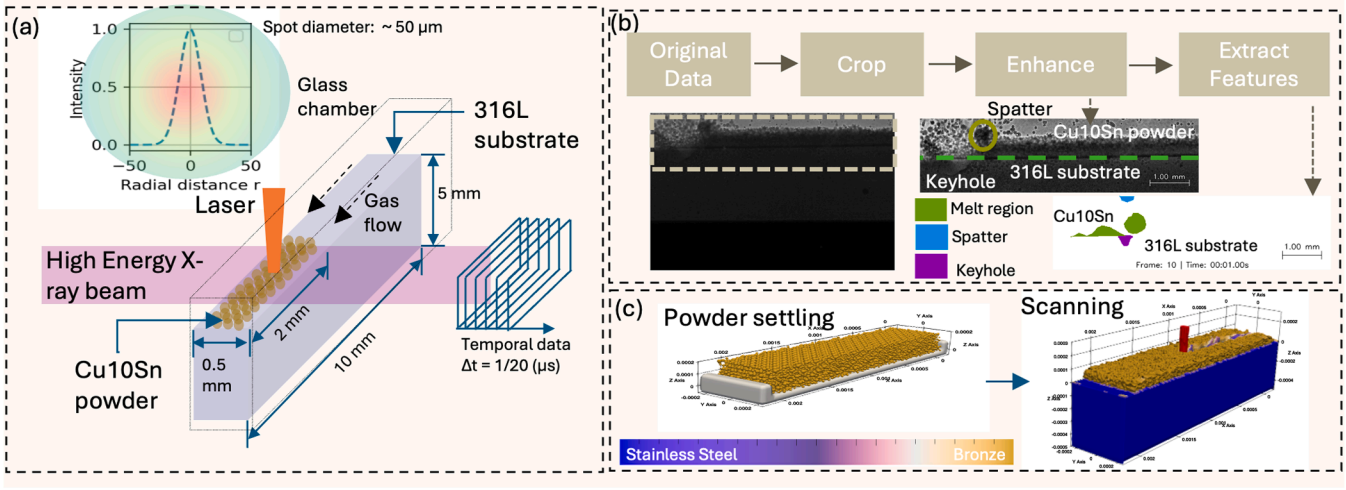


Fig. 2. (a) MM-LPBF set up within *In-situ* X-ray synchrotron operando set up, (b) Data-processing pipeline and (c) Computational numerical simulation set up.

in Supplemental Material S_A). When both materials coexist within a computational cell, local thermophysical properties, including density, viscosity, thermal conductivity, specific heat, and surface tension, are calculated as volume-fraction-weighted averages; all these properties are defined as temperature-dependent. The current simulation setup, however, does not account for any compositional effects of the materials, such as a liquid miscibility gap or atomic diffusion. Despite the limitations, the simulations can be qualitatively interpreted as all fluid properties, more importantly, the temperature-dependent surface tension effect (i.e., the Marangoni coefficient) for both materials are incorporated into the computational domain. As illustrated in Fig. 2C, bronze powder particles were deposited onto a stainless-steel layer using the discrete element method (DEM), and a multi-physics framework was employed to simulate laser-induced melting and fusion. The Volume-of-Fluid (VOF) method was used to track phase interfaces, while the governing continuity, momentum, and energy equations were solved within each computational cell [34,35]. The laser heat source model incorporated an angle-dependent Fresnel reflection formulation with up to 20 allowable reflections. Multiphase modeling, recoil pressure effects, and temperature-dependent thermal and fluid properties were included to accurately capture melt pool dynamics. Recoil pressure acting on the melt pool free surface was modeled by assuming a cylindrical vapor plume aligned with the laser beam axis. Within the plume region, each computational cell interacts with both the fluid and free surface, enabling coupled heat, mass, and momentum transfer. Equations (1) and (2) are used to compute evaporated mass and plume-induced force acting on the free surface which is critical for accurately capturing keyhole evolution during laser processing. In this formulation, Mdenotes the molecular weight of the vapor, R is the universal gas constant, and T_l, P_l, T_v, and P_v represent the liquidus and evaporation temperatures and pressures, respectively. The parameters $a_{vap}$and a_{con}correspond to the accommodation coefficients for evaporation and condensation. P_{recoil} represents the vapor pressure at the liquid free surface and defines the recoil pressure acting on the melt pool, which drives the plume-induced force (F_{plume}). The pressure magnification coefficient, m_p, was assumed to be unity in the present study.

$$m_{transfer} = \sqrt{\frac{M}{2\pi R}}\left(\frac{a_{vap}P_l^{sat}}{\sqrt{T_l}} - \frac{a_{con}P_v}{\sqrt{T_v}}\right) \quad (1)$$

$$F_{plume} = m_p . P_{recoil} \quad (2)$$

3. Results – spatter formation and implications in MM-LPBF

High-speed X-ray imaging of the MM-LPBF process revealed several dynamic phenomena in the melt pool. Two major processing events are identified: (1) spatter formation and (2) melt pool backfilling. The corresponding process videos are provided in the Supplemental Material: Video S-B presents the raw radiographic data, in which the overall process dynamics are visible, whereas Video S-C shows the extracted process features for enhanced observation of characteristic process signatures. Powder-type spatter formation was observed continuously throughout the laser scanning process, while a distinct droplet-type spatter event occurred intermittently (Time 2.60 s– 3.90 s), and melt pool backfilling was observed near the end of the scan track (Time 5.20s–6.00 s) and was accompanied by droplet spatter formation. Bronze is inherently challenging to process via laser-based manufacturing due to its high thermal conductivity and low laser absorptivity, which require higher energy densities to achieve complete melting during LPBF. When bronze is deposited onto stainless steel layers having comparatively lower thermal conductivity, heat dissipation from the melt pool is reduced. The resulting thermal accumulation enhances melt pool instability, thereby promoting spattering and backfilling. These phenomena are discussed in the following sections. Fig. 3 shows the instances of spatter formation. Both types of spatters - powder spatters and droplet spatters can be observed, which is consistent with other studies [47]. Powder spatters are highlighted in Fig. 3A, and the droplet spatter event can be seen in Fig. 3B.

3.1. Powder spatters

Powder spatters were observed continuously throughout the laser scanning process. In LPBF, the intense localized energy input generates metal vapor, which induces powder spatters by ejecting un-melted or partially melted particles from the front and sides of the melt track. Continuous powder spatter ejection during single-material LPBF has also been reported in previous studies [33,36,51]. The spatter trajectories highlighted in Fig. 3A indicate that particles are driven outward. At specific time instances (e.g., at time 4.50 s), some particles were observed to circulate behind the melt pool, suggesting the influence and directionality of the vapor plume and gas flow. Experiments were performed in a thin-walled carbon chamber filled with argon gas at atmospheric pressure (Fig. 2A). Although no externally controlled gas flow was introduced, metal vaporization during processing is expected to induce a local gas flow opposite to the scan direction, driven by pressure gradients associated with the vapor plume and the localized heating of the surrounding gas.

Powder spatter is detrimental to the LPBF process because it reduces the effective powder volume available for laser-material interaction [47]. In the context of MM-LPBF, where bronze powder is deposited onto stainless steel, powder spatter events can expose the underlying steel layers to direct laser irradiation. Due to the higher laser absorptivity and lower thermal conductivity of stainless steel relative to bronze, the effective energy input to the melt pool is intensified, potentially affecting melt pool stability. The keyhole observed in the present experiments exhibited highly dynamic behavior, and powder spatter may be a potential trigger for this instability. In addition to the ejection of unmelted powder particles, commonly referred to as cold ejections, particles entrained within the laser path may partially melt before being expelled by the vapor plume, a process known as hot ejections [47] or entrainment melting spatter [51]. Hot spatter particles are often larger and less dense, whereas cold spatter particles are comparatively smaller and denser. Hence, the cold and hot powder spatters can be distinguished by the pixel's greyscale values, but the particles' actual temperatures cannot be validated [47]. Furthermore, spatter particles may agglomerate following ejection, as illustrated by the representative subcategories highlighted in Fig. 3A. In the present study, powder spatters were observed in the range of 30 - 80 μm (±0.0062 μm), cold spatters were approximately <40 μm (±0.0062 μm).

3.2. Droplet spatters

Droplet spatter events were observed near the beginning of the laser scan path (see Supplemental Videos S-B and S-C). Fig. 3B illustrates one representative droplet spatter ejection event. The average size and

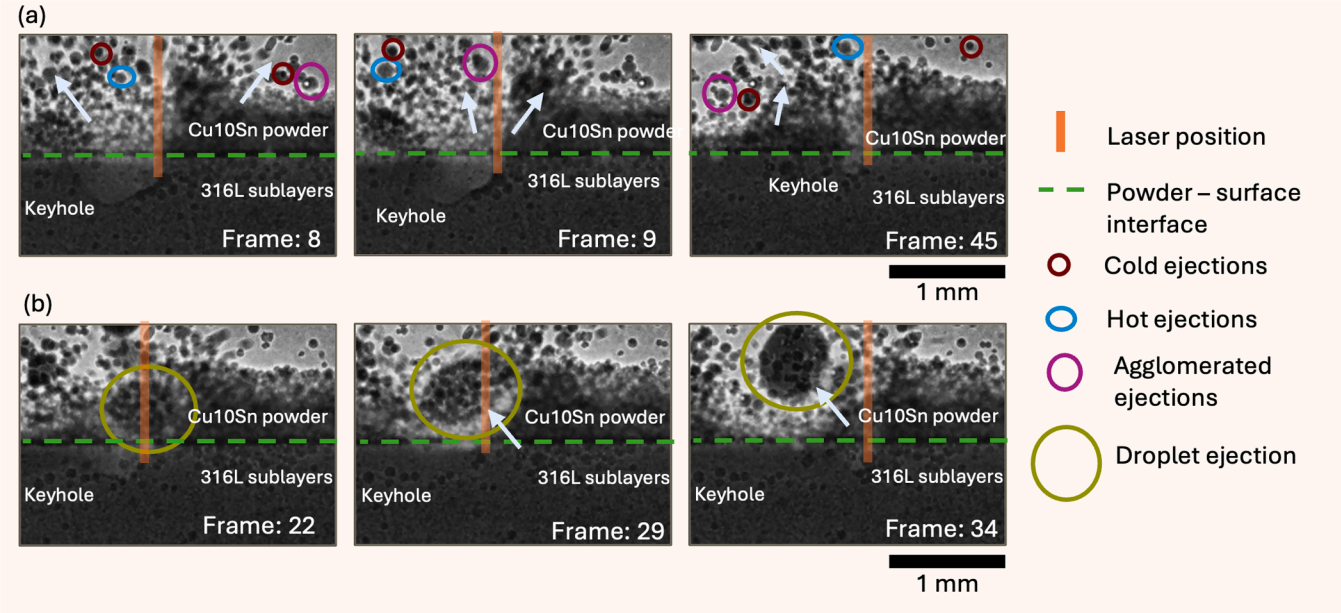


Fig. 3. Representative events of spatter formation and ejection in MM-LPBF: (a) powder spatter (b) Droplet spatter. **In-situ* observations show spattering events during laser scanning occurs continuously, and droplet spatters appear intermittently.

velocity of the observed droplet spatter were approximately 625 μm ±0.0062 μm and 0.6 m/s ±0.000175 m/s, respectively. Although the measured spatter velocity is lower than values commonly reported in previous studies [52], the droplet size observed in the present work is larger [33]. It should be noted that the experiments were conducted in an inert gas chamber at atmospheric pressure, with low-pressure gas flow due to process-induced metal vaporization. High-pressure gas flows (typically used in LPBF machines) can break up droplets and impart high velocity to the ejections. The material properties and processing parameters strongly influence spatter dynamics and ejection velocity.

In the present study, three distinct droplet spatter ejection events are clearly identified (see Supplemental Material S-B), while some smaller droplet spatter generation may be overlaid with the powder spatters that are difficult to distinguish. The formation of large droplet spatter can be attributed to high turbulence and excessive metal vaporization within the melt pool, caused by a mismatch between the processing regions of the two materials (i.e., bronze vs. steel). Recoil pressure applied by the metal vapor follows the ideal gas law (P = nRT/V, P – pressure, n – number of vapor particles (moles), R – gas constant, T – temperature, V-Volume), which states the linear relationship between the temperature and pressure. The steel absorbs and retains more energy in the melt pool, increasing its effective temperature and vaporizing excess bronze. Excessive metal vapor could destabilize the melt pool, leading to larger droplet spatters. On X-ray radiographs, distinguishing between the two materials is difficult because their X-ray attenuation coefficients are similar. To further analyze the interaction of the two materials in the melt pool, a CFD simulation is used. The results are discussed in the following section. Droplet spatters are detrimental to the process, as they lead to keyhole instability, potential pore formation, and uneven surface topography. Large droplets could shadow incoming laser energy, leading to uneven melting and a lack of fusion voids in the build [48]. In the context of MM-LPBF, droplet spatters can lead to a non-uniform material-mixing interface, potentially resulting in mechanical properties that deviate from the intended design. Droplet spatter could land on the MM-LPBF powder bed, causing chemical contamination and stochastic defect formation.

3.3. Melt pool back filling

Another notable phenomenon observed during laser scanning was melt pool backfilling, which has received comparatively limited attention in the additive manufacturing literature. In laser-based manufacturing processes, the flow behavior of molten metal is strongly governed by keyhole dynamics [41]. Localized high-energy laser irradiation induces intense metal vaporization, leading to the formation of a keyhole driven by recoil pressure at the vapor-liquid interface. Kouraytem et *al.* [53] discuss the keyhole dynamics and note that the backward flow of metal, mainly caused by thermocapillary action (or Marangoni effect) is highly dependent on keyhole morphology. Molten metal flows around the keyhole from both the side and bottom regions before flowing into the trailing melt pool. Accumulated molten metal at the tail of the melt pool tends to flow upward, forming a hump on the surface of the melt track.

In this work, when bronze is deposited on top of stainless steel 316 L layers, the melt pool contains both materials. Upward flow could carry stainless steel into the bronze region, inducing heterogeneous mixing. Fig. 4 shows the timesteps of the melt pool back-filling event from the *in-situ* X-ray radiation data. The molten metal initially flows up at the front of the keyhole, indicating a mechanism of droplet spatter formation similar to that previously discussed in Fig. 3B. The fluid lacks sufficient energy to be ejected and is pulled back by thermocapillary forces towards the tail of the melt pool. Backfilling creates an uneven surface topography by destabilizing the melt pool tail and forming hump-like features. As with droplet spatter, melt-pool backfilling could adversely affect the uniformity of material mixing in MM-LPBF.

3.4. Evolution of melt pool morphology

Fig. 5 presents the temporal evolution of keyhole dimensions extracted from the *in-situ* X-ray radiographs. The keyhole length fluctuates throughout the process, while the depth of the keyhole remains comparatively constant, except near spattering and melt-pool backfilling events. These results show unstable melt pool dynamics, as previously noted due to spattering events [33,48]. When molten material is ejected from the melt pool surface, the keyhole dimensions temporarily increase until subsequent material melting replenishes the melt pool volume. The measured melt pool geometry indicates operation in the conduction mode, with a depth-to-length ratio below 0.8; consequently, no keyhole-induced porosity was observed.

3.5. Numerical simulation of spattering in MM-LPBF

A CFD model was developed to interpret *in-situ* X-ray imaging observations by analyzing thermal gradients, laser-material interactions, and the mixing behavior between the two metals. Although spatter formation and melt pool backfilling were clearly observable in the X-ray radiographs, the individual materials could not be distinctly resolved because of their similar X-ray attenuation characteristics. Supplemental Videos S-D to S-F present the numerical simulation results for the deposition of bronze powder onto stainless steel 316 L. Specifically, Video S-D illustrates the evolution of the temperature distribution, Video S-E shows the predicted material-mixing behavior between the two metals, and Video S-F presents the molten-metal velocity field.

Droplet spatter events were clearly reproduced in the simulations, demonstrating the ability of the numerical model to capture process instabilities consistent with experimental observations.

It should be noted that the present simulation framework includes the following assumptions: (1) The movement of independent particles in the simulation space is restricted. Hence, the model does not account for the effects of powder spatters. (2) Effects of compositional variation on the melt pool dynamics are neglected, i.e., liquid miscibility gap or atomic diffusion phenomenon are not considered. Previous work has noted that powder spatters can lead to non-uniformity in powder layer thickness but have comparatively less pronounced effects on melt pool dynamics [51]. A liquid miscibility gap in the Fe-Cu system forms when the mixture undercools, becomes thermodynamically unstable, and separates into two liquids of different compositions [54]. The separation temperature depends on the material composition, which can affect the

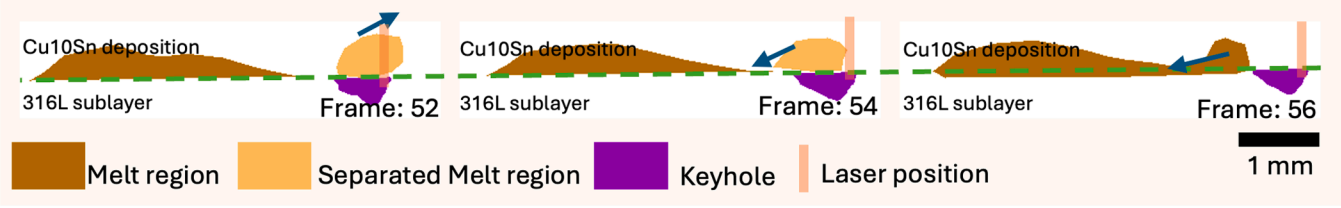


Fig. 4. Melt pool backfilling in MM-LPBF captured by *In-situ* X-ray radiation when bronze is deposited on stainless steel. *The X-ray radiation data is processed to extract features, melt regions and keyhole.

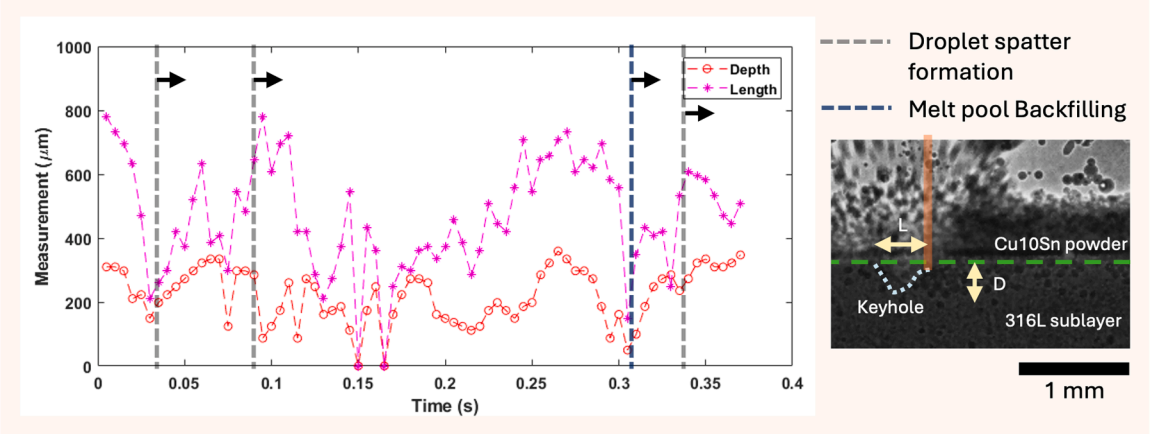


Fig. 5. Keyhole dynamics with the highlighted events of melt pool backfilling. *Keyhole morphology evolves during the laser-scanning, specifically following the spattering and melt pool backfilling events. The observed instability in keyhole indicates dynamic melt pool behavior because of spattering occurrence.

interfacial mixing region. But all melt-pool dynamic events occur above the liquidus temperature (>1700 K), where the mixture is not separated. In the current simulation setup, all material-specific fluid properties, especially Marangoni convection ($\frac{d\sigma}{dT}$, σ is surface tension in kg/s^2 and T is temperature in K), effects of both materials are incorporated. Hence, the simulation results can be interpreted qualitatively, while quantitative agreement would require incorporating more advanced physical modeling approaches, such as coupled discrete element method (DEM) simulations, and accounting for the effects of compositional gradients on surface tension. The previous simulation works in MM-LPBF with the same assumptions showed good agreement with the experimental results [19].

Fig. 6 illustrates the predicted material distribution after the laser scanning. The results show heterogeneous mixing regions between the two materials, corresponding to spattering events. These spatters are droplet ejections and mostly consist of bronze; they can be seen accumulating behind the laser scan path. When droplet spatter is ejected from the surface of the melt track, it draws 316 L stainless steel from the melt pool below the bronze toward the spatter. The bronze material flows back from the front and bottom of the keyhole, potentially forming a non-homogeneous sublayer. The overall mixing in the melt flow exhibits signs of Marangoni convection, while spatter ejections and melt pool backfilling introduce heterogeneous mixing regions (elongated and abrupt within the melt pool). Fig. 6 presents selected simulation time steps illustrating representative spatter-generation events. The first spatter event (Frame 13) demonstrates the formation and ejection of a large, non-spherical spatter particle composed of bronze, while a portion of the stainless-steel material is simultaneously entrained in the direction of ejection. This spatter originates from the rear wall of the keyhole, whereas subsequent spatter events are primarily initiated from the front region of the keyhole. The observed droplet spatters, which exhibited near-spherical morphologies *in-situ* studies, were not accurately reproduced in CFD simulations (e.g., spatter shapes). This discrepancy is due to the absence of explicit gas-phase flow modeling associated with metal vapor expansion and inert gas dynamics in the CFD setup. Although the simulation framework accounts for phase transformations and recoil pressure effects through a predefined vapor plume geometry and flow direction, the metal vapor transport was not explicitly resolved in these CFD analyses.

Fig. 7 shows the process instability predicted by the simulation

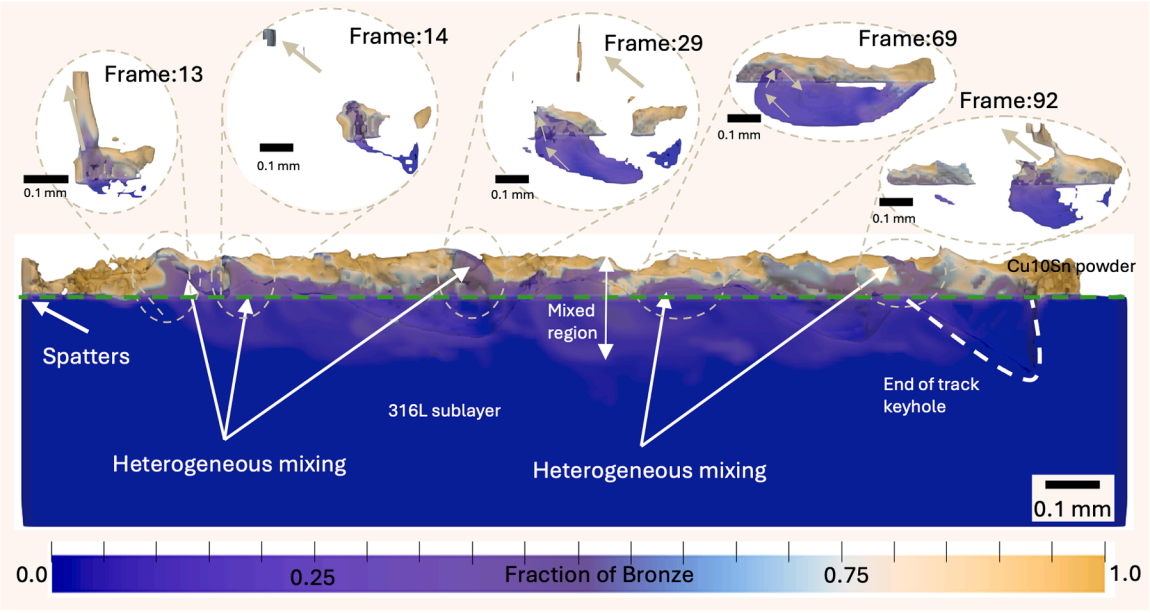


Fig. 6. Material mixing predicted by numerical simulation; highlighted frames show melt pool shapes that caused heterogeneous mixing. *Simulations predict the unstable processing dynamics including droplet spattering events. The spatter ejection contains bronze entraining subsurface molten steel into the bronze layer.

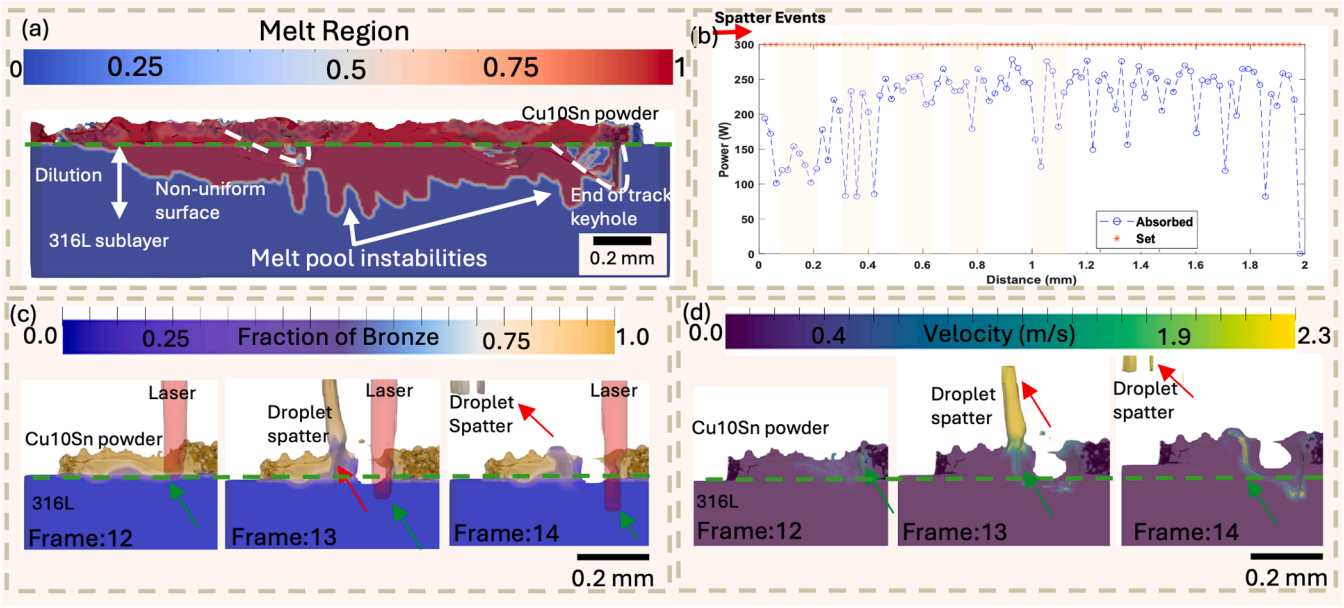


Fig. 7. Melt pool dynamics (a)Total melt region (b)Power vs distance; the highlighted regions are spattering events (c) Laser penetration during selected spattering events (d) predicted velocity of droplet spatters. *Simulations capture melt pool instabilities that lead to a non-uniform layer surface and fluctuations in melt pool depth.

predictions. Fig. 7A and B are the total melt region and variation in the absorbed power by the melt pool. Initially, when the laser interacts with the subsurface steel, fluid turbulence increases, and droplet spatter is ejected from the back of the melt pool (Fig. 7C and D). Laser scanning under the same processing conditions as single-material LPBF of bronze shows a stable melt pool evolution, with no spatter observed; supplemental S_G shows the single material (bronze powder and sublayers) simulation results. This suggests that the dynamic power absorptivity introduced by laser interaction with the subsurface steel is a critical cause of melt pool instability, leading to spatter generation, melt backfilling, and, consequently, heterogeneous mixing of the two materials [19,55].

3.6. Post-evolution of the laser track

Fig. 8A presents the cross-section of the MM-LPBF sample, which shows the non-uniform layer morphology and heterogeneous mixing regions, manifested as isolated stainless-steel islands within the bronze regions. The locations of these features closely correlate with the droplet spatter events observed in the *in situ* X-ray imaging data. The morphology of the isolated steel islands is consistent with the heterogeneous mixing regions predicted by the numerical simulations. This suggests that spatter formation is the primary mechanism driving the development of a heterogeneous mixing zone with elongated, abrupt material islands. It is postulated that melt pool instability promotes spatter generation, during which molten bronze is ejected from the melt pool surface while simultaneously entraining stainless steel from the underlying layers along the spatter trajectory. Because bronze and stainless steel are immiscible, the entrained mixture undergoes liquid-phase separation, forming distinct, isolated islands [56].

In addition to the isolated steel islands, heterogeneous mixing regions exhibit evidence of thermocapillary-driven flow associated with melt-pool backfilling. In these regions, stainless steel from the previous layers is transported upward into the melt pool, as predicted by the simulations. Overall, the experimental cross-sectional observations validate the numerical simulations and provide new fundamental insight into the interaction between dissimilar metals and the role of melt-pool instability in governing heterogeneous mixing behavior during MM-LPBF.

3.7. Process recommendations for copper – steel multi-material system

The present study investigates the detrimental effects of spattering, which may occur at higher energy densities. Compared to single-material processing of copper (or similar materials with relatively higher thermal conductivity and lower laser absorptivity), the process energy density should be reduced when processing over a steel sublayer (or similar materials with relatively lower thermal conductivity and higher laser absorptivity) by either reducing power and/or increasing scan speed. Previous work by Griffis et al. [4] and Mao et al. [19] found that recoil pressure and Marangoni convection play significant roles in governing material mixing during MM-LPBF of the copper–steel system. It was noted that increasing the energy density improves uniform mixing of materials and a gradual transition in material properties across the multi-material interface. However, spattering events that may emerge at higher energy densities were not reported. Future studies could focus on

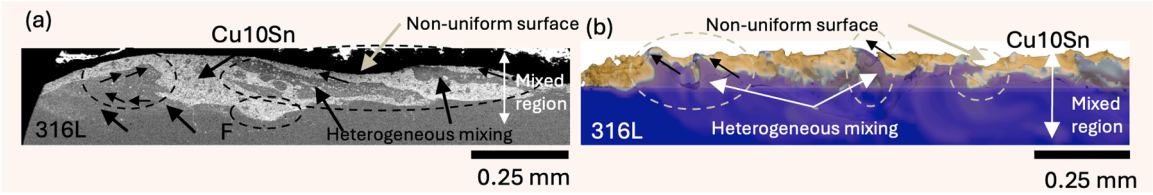


Fig. 8. Post evolution of sample (a) cross-section of the sample (b) Numerical simulation results. *The heterogeneous mixing predicted by the simulation shows good agreement with experimental results; subsurface steel is entrained in the bronze layer along the same trajectory as droplet spatters.

determining optimal processing conditions near the interface region and on additional interface orientation conditions.

4. Conclusion

This study investigated the influence of spatter formation mechanisms at a dissimilar-material interface during MM-LPBF through a combination of *in situ* X-ray imaging and numerical simulation. The results provide fundamental insight into the material-mixing behavior induced by melt-pool instability, particularly under spattering conditions, and demonstrate that differences in laser-absorptivity and thermal properties between the interacting materials can develop unsteady melt-pool flow, which promotes spatter formation. During droplet spatter ejection, stainless steel from underlying layers is entrained along the spatter trajectory, resulting in elongated, isolated stainless-steel islands that contribute to heterogeneous material mixing. In addition to spatter formation, melt pool backfilling was observed, where molten material flows toward the trailing region of the melt pool under the influence of Marangoni convection from the side and bottom regions of the keyhole. This also contributes to heterogeneous mixing behavior that could lead to interfacial cracking and spatially varying mechanical properties in MM-LPBF components. Finally, the resulting nonuniform surface topography can adversely affect subsequent layer deposition, potentially leading to processing defects such as lack-of-fusion porosity. Spatter deposition onto adjacent regions of differing composition may also locally alter the chemical composition of the build. While previous studies have suggested using high energy densities to improve interfacial bonding and compositional homogeneity during copper-alloy deposition onto Stainless-steel layers, the present work highlights the detrimental effects of excessive energy input and the resulting melt-pool instabilities. Future research should focus on investigating melt-pool dynamics across different build orientations and on identifying the fundamental mechanisms governing material mixing and defect formation in MM-LPBF systems.

CRediT authorship contribution statement

Aishwarya Deshpande: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Ziheng Wu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eric Elton:** Conceptualization. **Alan Burl:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation. **Harsh Mali:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation. **Gabe Guss:** Writing – review & editing, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Justin Patridge:** Visualization, Validation, Resources, Methodology, Investigation, Data curation. **Michael Troksa:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Laura Marquez:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Phillip Depond:** Writing – original draft, Validation, Methodology, Investigation, Data curation. **Thejaswi Tumkur:** Writing – original draft, Validation, Methodology, Investigation, Data curation. **Benjamin Ellyson:** Writing – original draft, Validation, Methodology, Investigation, Data curation. **Steven Van Petegem:** Visualization, Validation, Resources, Methodology, Data curation, Conceptualization. **Yunhui Chen:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alexander Rack:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Data curation. **Guha Manogharan:** Writing – review & editing, Writing – original draft, Supervision, Resources, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344 and was supported by the LLNL-LDRD Program under Project No 22-ER-018. LLNL-JRNL- 201928. We acknowledge the European Synchrotron Radiation Facility (ESRF) for provision of synchrotron radiation facilities under proposal MA-6162 and on beamline ID19. The authors would also like to thank Prof. Anthony Rollett for his assistance with the beamline proposal and for his time SVP is grateful for financial support from the Swiss National Science Foundation (SNSF) (Grant No. CRSII5_193799).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.addlet.2026.100409.

Data availability

Data will be made available on request.

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